Activity 5 Report: Database Maintenance and Monitoring

COMP 018 Database Administration | Project tested: PUP-SPOT Campus Lost and Found (PHP + PostgreSQL) | Date of testing: 6 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	PUP-SPOT Campus Lost and Found
Owning group	Group 05 (grp05-dba)
Repository	https://github.com/grp05-dba/grp05_lostandfound
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a5_health.txt, a5_logs.txt, a5_maintenance.txt

Run on 6 November 2026 at 23:22 (Asia/Manila), after the application checks of Activity 3. Results come from the saved files.

2. Health and activity

Check	Result
Version	PostgreSQL 18.6, x86_64 Windows
Database size	8566 kB
Connections	1 backend (the monitoring session). max_connections is 100
Transactions	462 commits, 16 rollbacks (the rollbacks come from the access tests)
Cache	40,248 buffer hits against 10 reads, about 99.98% hit ratio
Deadlocks and temp files	0 and 0
Autovacuum	on. No table had been vacuumed or analyzed before this run
Rows	categories 11, claim_images 0, claims 4, item_images 0, items 7, lost_report_images 1, lost_reports 9, potential_matches 7

Dead tuples before maintenance: only claims, with 2 (from the claim approval made in the tests). Below the autovacuum threshold.

3. Logging

Setting	Value	Comment
log_statement	none	No statement logging
log_min_duration_statement	-1	Slow queries are not logged. Recommend 500 ms
Log folder	C:/Program Files/PostgreSQL/18/data/log	Readable

4. Log findings

Source	Entry	Explanation
PostgreSQL 23:15:21	role lostandfound_public (and lostandfound_app) is not permitted to log in	Deliberate: each role was disabled for one minute to prove which role each server uses (Activity 3)
PostgreSQL 23:21:54 to 23:21:56	permission denied for table lost_reports, claims, items, claim_images; for schema public; to create role; must be owner; password authentication failed	The deliberate access tests of Activity 4
PostgreSQL 23:21:55 and 23:22:22	permission denied to vacuum and analyze "items"	lostandfound_app trying maintenance (see section 5)
PHP, public and admin servers	Undefined array key EMAILJS_* and TEXTBEE_* on every report, claim and item write	Missing optional keys (Activity 3, issue 1)
PHP, public server	Uncaught TypeError in publicController.php line 40	The empty-body request test (Activity 3, issue 2)
PostgreSQL, earlier today	Errors and failed logins from other projects tested on the same server	Not part of this project

No unexplained errors from the application.

5. Maintenance performed

Action	Result
ANALYZE (all tables)	Completed. last_analyze set on all 8 tables
VACUUM (VERBOSE, ANALYZE) public.potential_matches	7 live rows, 0 removed, 1 page set all-visible, 0.00 s
VACUUM and ANALYZE as lostandfound_app	Skipped with "permission denied". The project defines no maintenance role

6. Query plan

Query: unclaimed items whose title contains "umbrella" and location contains "library", joined to categories. EXPLAIN (ANALYZE, BUFFERS) showed a hash join with sequential scans, planning time 2.09 ms and execution time 0.05 ms, 1 row returned. With 7 rows a sequential scan is the right plan.

The database has 10 indexes, all primary keys and the two unique code columns. The search uses ILIKE with a leading wildcard, which an ordinary index cannot serve, and foreign key columns used in joins have no index: items.category_id, claims.item_id, potential_matches.report_id and item_id, and the three image tables.

7. Recommendations

- Create a maintenance role with pg_maintain and use it for scheduled VACUUM and ANALYZE.
- Set log_min_duration_statement to a value such as 500 ms and keep log_statement at ddl.
- Add indexes on the foreign key columns above, and a pg_trgm GIN index on items.title and location_found if the catalog grows.
- Turn PHP display_errors off and log to a file so warnings do not go to users.
- Monitor dead tuples and give a monitoring role pg_monitor.

8. Feedback sent to owners

Sent on 6 November 2026: no database health problems found. Recommended a maintenance role, slow-query logging, foreign key indexes and PHP error handling.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own server, and every result in this report comes from the saved output files.